

Logistic Regression (Binary classification) ←

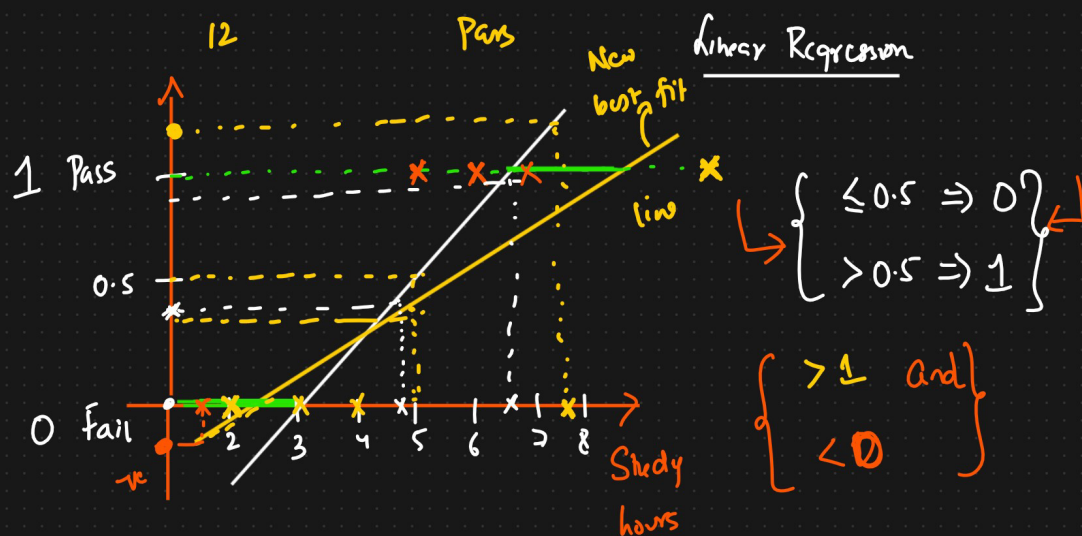
<u>Dataset</u>	<u>Pass/Fail</u>
Study hours	O/p {Binary categories}

2	Fail
3	Fail
4	Fail
5	Pass
6	Pass
7	Pass
12	Pass



Logistic Regression

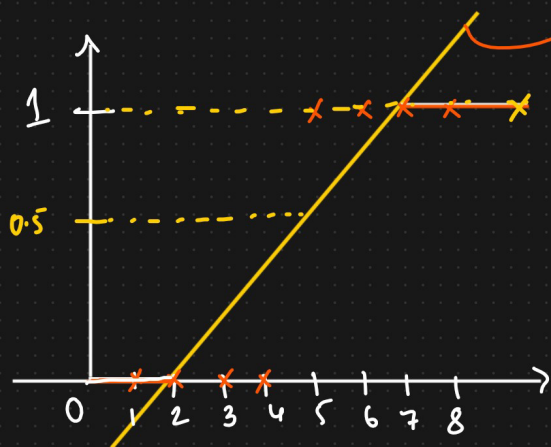
0 to 1



Why we cannot use Linear Regression for Classification?

- ① Outlier {Best fit line change}
- ② > 1 and < 0 {Squash line}

How Logistic Regression Solves Classification Problem

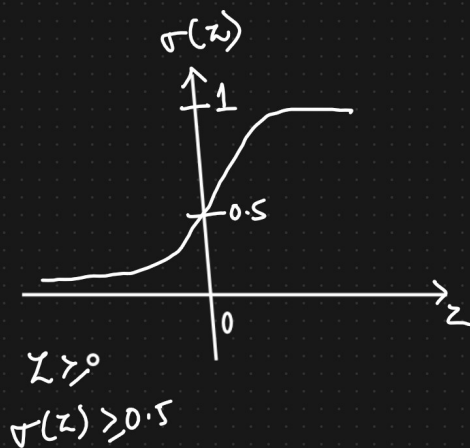


$$h_0(x) = \theta_0 + \theta_1 x_1$$

↓
Best fit line

Sigmoid Activation

$$\frac{1}{1 + e^{-z}}$$



$$h_0(x) = \sigma(\theta_0 + \theta_1 x_1)$$

$$\sigma = \frac{1}{1 + e^{-z}}$$

$$= \sigma(z)$$

$$h_0(x) = \frac{1}{1 + e^{-z}}$$

⇒ Logistic Regression hypothesis

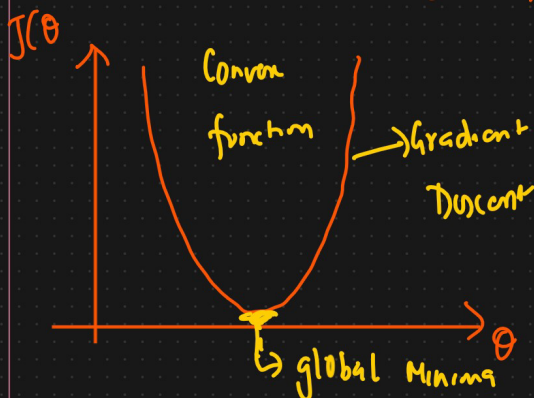
$$z = \theta_0 + \theta_1 x_1$$

Linear Regression Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

$$h_0(x) = \theta_0 + \theta_1 x$$

↓
Convex function



Logistic Regression Cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

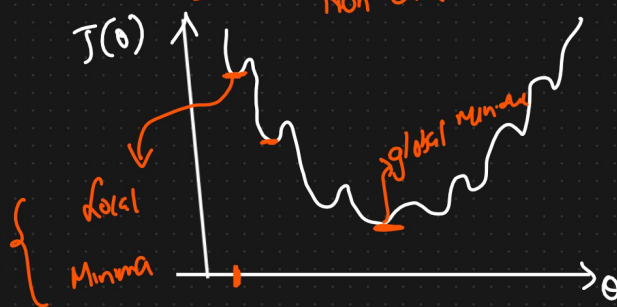
Non Convex function

$$h_0(x) = \frac{1}{1 + e^{-z}}$$

$$z = \theta_0 + \theta_1 x_1$$

Sigmoid

Non Convex



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \underbrace{(h_{\theta}(x)^{(i)} - y^{(i)})^2}_{\substack{\text{fits denote} \\ \text{Cost}(h_{\theta}(x)^{(i)}, y^{(i)})}}$$

$$h_{\theta}(x)^{(i)} = \frac{1}{1+e^{-z}} \quad z = \theta_0 + \theta_1 x_i$$

↓

fits denote $\text{Cost}(h_{\theta}(x)^{(i)}, y^{(i)})$

{ log loss }

$$\text{Cost}(h_{\theta}(x)^{(i)}, y^{(i)}) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y=1 \\ -\log(1-h_{\theta}(x)) & \text{if } y=0 \end{cases}$$

↓ convex function

$$\text{Cost}(h_{\theta}(x)^{(i)}, y^{(i)}) = -y \log(h_{\theta}(x)) - (1-y) \log(1-h_{\theta}(x))$$

$$J(\theta_0, \theta_1) = -\frac{1}{2m} \sum_{i=1}^m (y^{(i)} \log(h_{\theta}(x)^{(i)}) - (1-y^{(i)}) \log(1-h_{\theta}(x)^{(i)}))$$

Minimize Cost function $J(\theta_0, \theta_1)$ by changing

θ_0 & θ_1

Convergence Algorithm

Repeat

{

$j=0$ and 1

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}

Performance Metrics, Accuracy, Precision, Recall And F-Beta

Topics to be covered

- ① Confusion matrix
- ② Accuracy
- ③ Precision
- ④ Recall
- ⑤ F-Beta Score



Logistic Regression

R squared
Adjusted R squared

① Confusion Matrix

			Actual Values
	1	0	
1	3	2	
0	1	1	

Dataset		o/p	pred by model
x_1	x_2	y	$\hat{y}$
-	-	0	1
-	-	1	1
-	-	0	0
-	-	1	1
-	-	1	1
-	-	0	1
-	-	1	0

Predicted values

	1	0	Actual
1	TP	FP	
0	FN	TN	

$$\begin{aligned}
 \text{Accuracy} &= \frac{TP + TN}{TP + FP + FN + TN} \\
 &= \frac{3 + 1}{3 + 2 + 1 + 1} \\
 &= \frac{4}{7}
 \end{aligned}$$

② Dataset Binary classification

$\rightarrow$ 1000 datapoints $\left\{ \begin{array}{l} 900 \rightarrow 1 \\ 100 \rightarrow 0 \end{array} \right\}$ Imbalanced Dataset
 $\downarrow$
 90% Accuracy

④ Precision = $\frac{TP}{TP+FP}$ } Out of all the actual value
how many are correctly predicted

	1	0	Actual
1	TP	FP	
0	FN	TN	
Predicted			

② Recall = $\frac{TP}{TP+FN}$ } Out of all the predicted value
how many are correctly predicted

Usecase ①

Spam classification

Mail → Spam } Good
Model → Spam }

	Spam	Not Spam	
	1	0	Actual
Spam	TP	FP ✓	
Not Spam	FN	TN	

Mail → Not Spam } Blunder
Model → Spam }

$$\text{Precision} = \frac{TP}{TP+FP}$$

Usecase 2

To predict whether person has diabetes or not

✓ Truth → diabetes }
✓ Model → Doesn't diabetes } Blunder

	Diab	No Diab	
Diab	TP	FP	
No Diab	FN	TN	

$$\text{Recall} = \frac{TP}{TP+FN}$$

Truth → diabetes } good
Model → " }

usecase of diseases

$\left. \begin{array}{l} \text{Ternn} \rightarrow \text{Not diabetes} \\ \text{Model} \rightarrow \text{Diabetes} \end{array} \right\} \Rightarrow \begin{array}{c} \text{2nd opinion} \\ \Downarrow \\ \text{check} \end{array}$

Assignment

④ Tomorrow the stock market will crash or not

Reducing FPL or FN↓

④ F-Beta Score $\div (1 + \beta^2) \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

① If FP & FN are both important

$$\beta = 1$$

$$F1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

② If FP is more important FN

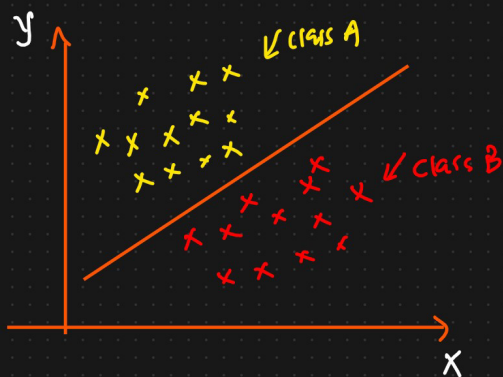
$$\beta = 0.5$$

$$F_{0.5} \text{ score} = \frac{(1 + 0.25) \cdot \frac{P \times R}{P + R}}{2}$$

③ gf $FN \gg FP$

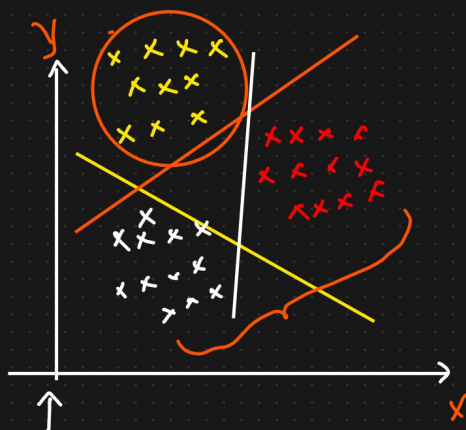
$$\beta = 2$$

Logistic Regression (One Versus Rest)



① Binary Classification

One Versus One



Multiclass Classification {Logistic Regression}

$\rightarrow M_1 \rightarrow$ Binary classification
 $\rightarrow M_2 \rightarrow$ Binary classification
 $\rightarrow M_3 \rightarrow$ Binary classification

One Versus Rest (OvR) $\rightarrow$ Logistic Regression

f_1	f_2	f_3	$\downarrow 0_1, 0_2, 0_3$	O/p	$\boxed{0_1}$	0_2	$\boxed{0_3}$
-	-	-	0_1	1	0	0	
-	-	-	0_2	0	1	0	
-	-	-	0_3	0	0	1	
-	-	-	0_1	1	0	0	
-	-	-	0_3	0	0	1	
-	-	-	0_2	0	1	0	

Model
 $\uparrow$
 $M_1 \leftarrow \text{I/P } \{f_1, f_2, f_3\} \text{ O/p } \{0_1\}$
 $M_2 \leftarrow \text{I/P } \{f_1, f_2, f_3\} \text{ O/p } \{0_2\}$
 $M_3 \leftarrow \text{I/P } \{f_1, f_2, f_3\} \text{ O/p } \{0_3\}$

$0.55 \rightarrow \text{O/p } = \boxed{0_3} \rightarrow \underline{\text{Category 3}}$

New Test Data $\rightarrow$
 $\left\{ \begin{array}{l} M_1 \rightarrow 0.25 \checkmark \\ M_2 \rightarrow 0.20 \checkmark \\ M_3 \rightarrow 0.55 \checkmark \end{array} \right.$
 $\uparrow \quad \uparrow \quad \uparrow$
 $M_1 \quad M_2 \quad \boxed{M_3}$
 $\uparrow$
 $\underline{\underline{\text{O/p}}}$